Tectonophysics, 100 (1983) 241–256
Elsevier Science Publishers B.V., Amsterdam – Printed in The Netherlands

PALEOMAGNETISM OF THE ESAYOO FORMATION (PERMIAN) OF NORTHERN ELLESMERE ISLAND: POSSIBLE CLUE TO THE SOLUTION OF THE NARES STRAIT DILEMMA

P.J. WYNNE [1], E. IRVING [1] and K. OSADETZ [2]

[1] *Pacific Geoscience Centre, Box 6000, Sidney, B.C. V8L 4B2 (Canada)*

[2] *Institute for Sedimentary and Petroleum Geology, 3303-33rd St. N.W., Calgary, Alta. T2L 2A1 (Canada)*

(Received May 5, 1983; accepted June 10, 1983)

ABSTRACT

Wynne, P.J., Irving, E. and Osadetz, K., 1983. Paleomagnetism of the Esayoo Formation (Permian) of northern Ellesmere Island: possible clue to the solution of the Nares Strait dilemma. In: M. Friedman and M.N. Toksöz (Editors), Continental Tectonics: Structure, Kinematics and Dynamics. *Tectonophysics*, 100: 241–256.

The Esayoo Formation (Lower Permian) of northern Ellesmere Island (81°N, 82°W) has been sampled at two localities (twelve sites, 88 cores studied) in the Early Tertiary (Eurekan orogeny) fold and thrust belt of the northern Sverdrup Basin, northern Ellesmere Island. The directions of remanent magnetization at the two localities are in excellent agreement after correction for geological tilt. Their mean is 126.2°, −61.8° (B (sites) = 12, $k = 105$, $\alpha_{95} = 4.3°$) and the corresponding paleopole is 48.1°N, 159.2°E ($K = 52$, $A_{95} = 6.1°$). Comparison with Early Permian reference paleopoles for North America shows that the formation of overthrust sheets has produced no paleomagnetically detectable displacement in a paleolatitudinal sense as would be inferred from the geology, but there is a significant anticlockwise rotation of $36 \pm 8°$. This rotation is the first to be observed in the Canadian Arctic Islands. Geological evidence allows no more than about 50 km of lateral motion along Nares Strait between Ellesmere Island and Greenland. This falls far short of the 300 km of relative motion required by the marine geophysical evidence for the opening of the Labrador Sea and Baffin Bay. This discrepancy between the geological and geophysical evidence is the Nares Strait paradox. Evidently relative motion between Greenland and Ellesmere Island is not restricted to Nares Strait itself and must be more generally distributed across Ellesmere Island. We suggest that the 36° rotation observed is a manifestation of this distributed strain, and therefore provides a clue to the solution of the Nares Strait paradox.

GEOLOGY

The Esayoo Formation is a Lower Permian volcanic unit which extends over approximately 24,000 km^2 in northwestern Ellesmere and northern Axel Heiberg

Contribution of the Earth Physics Branch no. 1075.

0040-1951/83/$03.00 © 1983 Elsevier Science Publishers B.V.

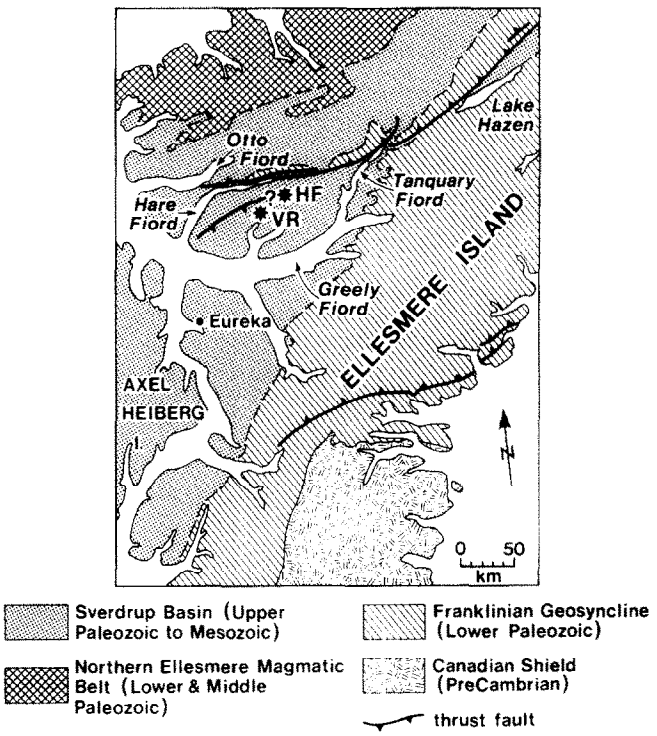


Fig. 1. Locality map showing the major geological units and sampling localities *HF* (Hare Fiord), and *VR* (Van Royen Ridge).

Islands. Measured thicknesses range from 37 m, on the north coast of Axel Heiberg Island to 297 m in the Van Royen Ridges of Ellesmere Island. It is composed of flows, 4–7 m thick, of reddish purple and green, fine grained basalt with subordinate basaltic agglomerate. Individual flows are dense and massive at the base, grading upward into amygdaloidal and vesicular lava. Nodular red chert beds at the tops of some flows are the only known interflow sediments. The flows were probably extruded quickly over a surface of little relief at or near sea-level.

The basal contact with the underlying Nansen Formation and the top contact with both the Degerbols and Trold Fiord formations in the Van Royen Ridge are sharp with minor irregularities. On northern Axel Heiberg Island the Esayoo Formation is overlain by shales of the Van Hauen Formation. The underlying Nansen Formation is Early Permian in age (Thorsteinsson, 1974). The overlying Degerbols and Trold Fiord formations are Late Permian and the Van Hauen Formation is Early Permian (Thorsteinsson, 1974). Hence the Esayoo Formation is Early Permian in age.

Although the stratigraphic relationships are well studied (Thorsteinsson, 1974) the structure is not well known. Low angle thrusts dipping both north and south are present.

ESAYOO FORMATION

The Esayoo Formation has been sampled at two localities 32 km apart. At Van Royen Ridge eight flows were sampled spaced through 297 m stratigraphically. At Hare Fiord five flows were sampled through 70 m stratigraphically. Between five and nine separately oriented cores were collected from each flow. From each core one or two specimens were studied. A place of collection is referred to as a site.

Detailed thermal and alternating field (AF) demagnetization was carried out on several specimens from each site. Except at collecting site 17 (described below) all specimens showed simple behaviour. They have a single predominant magnetization with reversed polarity. A typical example is shown in Figs. 2 and 3. AF demagnetization up to 100 mT shows a steady decrease in intensity with little or no change in direction. Most of the magnetization has remanent coercivities in excess of 100 mT. In some specimens there is a small initial increase in intensity. The component being removed during this increase has not been accurately estimated; it could be a small

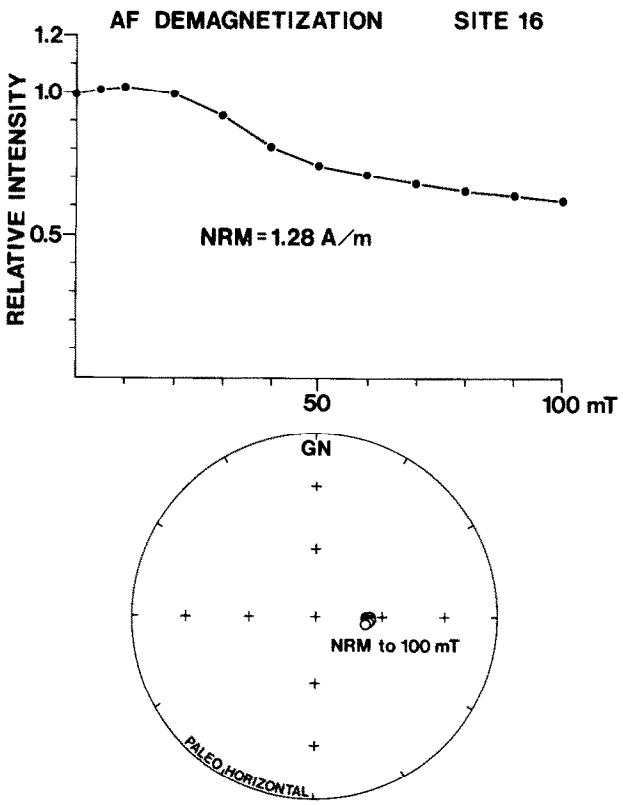


Fig. 2. AF demagnetization of massive Esayoo basalt from site 16: single component magnetization. Intensity changes and orthogonal plots above, direction changes below. Open (solid) symbols denote negative (positive) inclinations.

component aligned along the present earth's field (PEF) or it could be a small antiparallel component acquired in hematite during repeated cycling during Quaternary glaciations through the Morin transition (−15°C) comparable to that observed in laboratory experiments by Dankers (1983). During thermal demagnetization there is again a small increase in intensity owing to removal of a very small component in the direction of the PEF followed by a regular decrease from 450° to 675°C above which the magnetization becomes internally random. The directions remain constant throughout, and the excellent linear decay to the origin of the orthogonal plots shows that essentially only one component is present. Except for site 17, this behaviour is typical of the Esayoo Formation; the unblocking temperatures are distributed between about 450°–675°C and the coercivity of remanence commonly exceeds 100 mT. The fact that a substantial portion of the magnetization has unblocking temperatures in the range characteristic for pure magnetite (500°–580°C) would indicate that this mineral is present, but the high coercivity

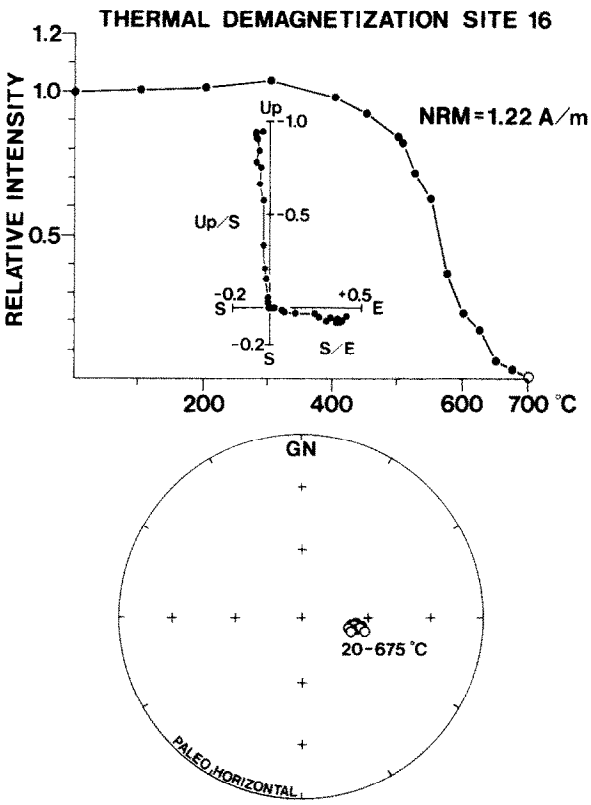


Fig. 3. Thermal demagnetization of massive Esayoo basalt from site 16: single component magnetization. Intensity changes and orthogonal plots above, directions below. The open symbol on the decay curve indicates that the direction is scattered.

suggests that hematite not magnetite is the major contributor to the remanence. This is born out by studies of polished sections from each site. Hematite is ubiquitous both as disseminated crystals, generally elongated ranging up to 100 μm and down to less than 1 μm in length, and as rims and fracture-fillings of olivine and pyroxene crystals. Only in two sections small (10 μm) discrete grains of magnetite were observed after a considerable search; it is a rare mineral in these rocks. Hence although magnetite could carry some of the remanence, hematite is the main source and the spread in unblocking temperatures is presumably caused by the presence in it of variable amounts of impurities (probably ilmenite) and by the wide range of grain-sizes present.

TABLE I

Results by sites

Site	S	N	M	D	I	k	α_{95}	Treatment
12	5	6	0.28	138	−62	86	7	60 mT, 550°C
13	9	11	1.14	122	−64	192	3	60 mT, 550°C
14	9	10	0.15	148	−55	51	7	70 mT, 550°C
15	6	7	0.66	132	−61	131	5	70 mT, 550°C
16	5	6	1.37	101	−68	133	6	70 mT, 550°C
17MR	3	5	0.06	121	−68	132	7	400–550°C
17MN	3	5	0.06	294	66	161	6	600–670°C
17AR	3	6	0.50	136	−66	72	8	70 mT, 550°C
17M+A	6	16	0.23	125	−66	94	4	
18	6	7	0.59	124	−59	54	8	60 mT, 550°C
19	6	11	2.16	035	83	22	10	70 mT, 550°C
20D	5	10	5.07	352	81	58	6	30 mT, 550°C
21	5	6	3.74	118	−60	596	3	70 mT, 550°C
22	5	5	1.09	113	−65	863	3	60 mT, 550°C
23	5	6	0.95	118	−59	147	6	70 mT, 550°C
24	5	6	0.38	148	−62	37	11	60 mT, 550°C
25	5	7	0.22	115	−52	184	5	60 mT, 550°C

Notes: The collecting sites have field accession numbers that generally have no stratigraphic significance. Site 20D is from the dolerite sill, the rest are from the Esayoo Formation. S is the number of cores and N the number of specimens used in the analysis. Of the 94 specimens studied from the Esayoo Formation 89 had a dominant component directed to the SE with negative inclination, five specimens (from site 17) contained two antiparallel components, eleven specimens (from site 19) had single component magnetization directed downwards and parallel to the dolerite (site 20D), and three specimens (from sites 12, 14, 22) had magnetizations that could not be satisfactorily resolved. M is the magnetization in A/m. D, I are the declination and inclination corrected for tilting; attitudes are 63° at 163° for sites 12 to 20, and 29° at 225° for sites 21 to 25. k and α_{95} are Fisher's precision and circle of confidence. In the last column the treatment is specified. At site 17 massive lava contains two antiparallel magnetizations; the reversed component (17MR) is obtained by calculating the magnetization removed between the 400 and 550°C heating steps, and the normal component (17MN) is obtained by cleaning in the range 600 to 670°C. Amygdaloidal lava at site 17 (17AR) has a single component. Entry 17M+A is the average of the results for the massive and amygdaloidal specimens irrespective of sign and this is used as the mean for site 17 in the analysis of Table II.

At site 17 purple massive and amygdaloidal basalts occur. The amygdaloidal basalt has properties similar to those just described. The massive basalt is quite different. It has two, oppositely-directed magnetizations. Up to 400°C there is little change in intensity, but between 450° and 600°C there is a spectacular increase in intensity (Fig. 4). The magnetization removed between these temperature has an antiparallel direction. Finally between 650° and 680°C there is a sudden drop in intensity. Despite these very large changes in intensity, the direction remains constant throughout the entire temperature range showing that the normally magnetized high-temperature component is much the larger. The averages of the high-temperature (17MN, Table I) and lower temperature (17MR, Table I obtained by vector subtraction) magnetizations in all massive basalt specimens at site 17 are

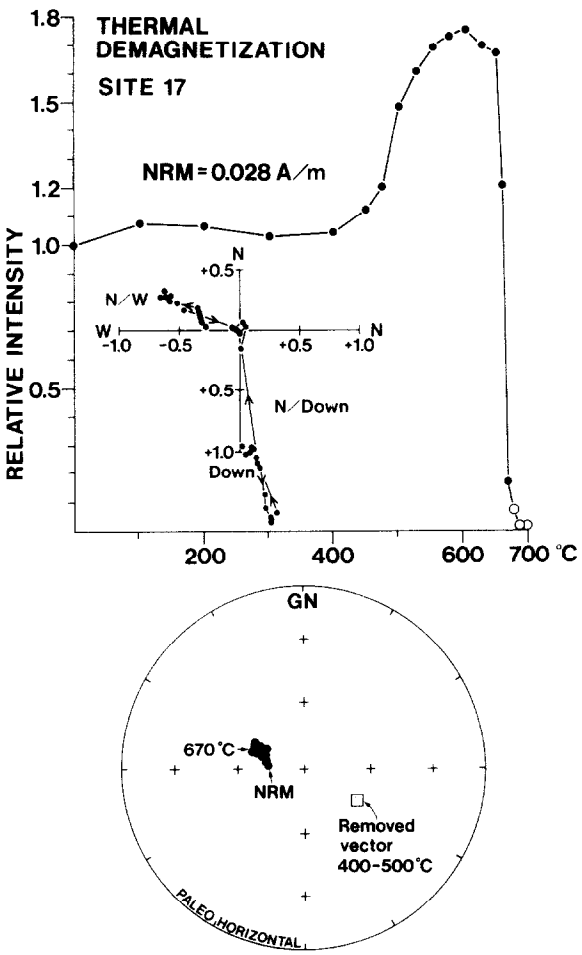


Fig. 4. Thermal demagnetization of massive Esayoo basalt from site 17: two component magnetization. Intensity changes and orthogonal plots above, directions below.

accurately antiparallel. The AF demagnetization results show no evidence of two magnetizations. There is a small, steady increase in intensity up to 100 mT (Fig. 5). In polished section the amygdaloidal basalt contains a small amount of finely crystalline hematite associated with pyroxene. Hematite is more abundant in the massive basalt where it occurs as rims around pyroxene and as thin elongate crystals approximately 100×10 μm. Evidently magnetite is rare or absent and both magnetizations, despite their spread of unblocking temperatures, must be carried by hematite. The two components could reflect a reversal of the field or a self-reversal.

All specimens from site 17 have been demagnetized thermally in detail as shown in Figs. 4 and 5 in order to separate the components accurately. At other sites this is unnecessary because the behaviour is much simpler, and there is a wide range of temperatures and alternating fields in which the orthogonal plots show linear decay to the origin (i.e. stable "end-points") so that the choice of cleaning treatment is not critical. All specimens were demagnetized at two or more intervals in the range of linear decay (as judged from test specimens) to confirm the repetition of directions. The cleaning procedures used are listed in Table I. The directions of magnetization

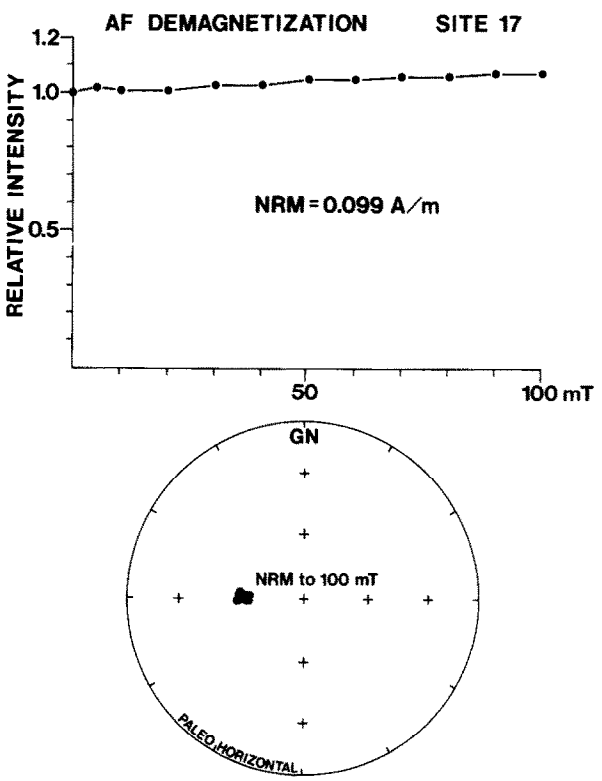


Fig. 5. AF demagnetization of massive Esayoo basalt from site 17: two component magnetization. Intensity changes above, directions below.

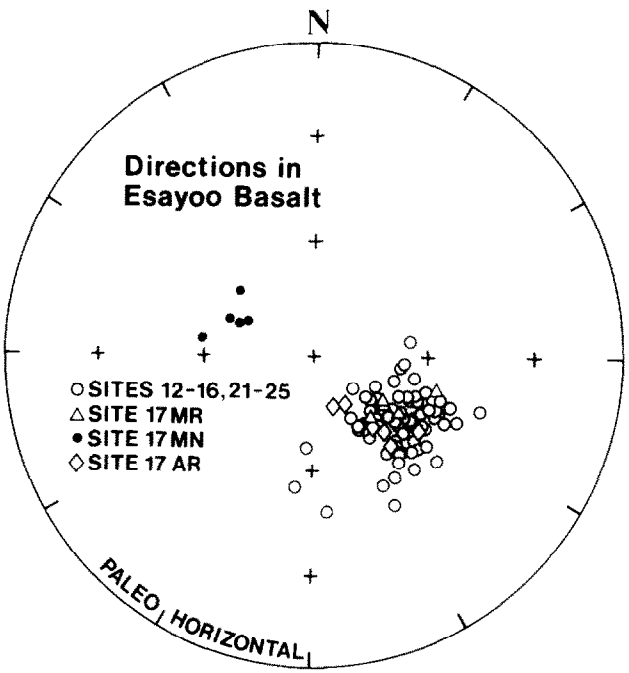


Fig. 6. Directions of magnetization in Esayoo basalt after cleaning and correction for tilt.

of all the specimens after cleaning and correction for bedding tilt are shown in Fig. 6. With the exception of the specimens from the massive purple basalt at site 17 they are directed to the southeast with negative inclinations.

DOLERITE SILL AND REMAGNETIZED ESAYOO BASALT

A dolerite sill 105 m in thickness was sampled at site 20. It has a simple magnetization. AF demagnetization yields consistent directions which become internally random above 80 mT (Fig. 7). Thermal demagnetization yields a near-perfect square-shouldered curve with unblocking temperatures between 500° and 575°C with random directions above that temperature (Fig. 8). Polished sections of the dolerite show large grains of magnetite (200 μm) with ilmenite lamella. After cleaning, all specimens have a well-grouped, near-vertical, downward direction, as would be expected for Cretaceous rocks (Fig. 9, Table I). Samples of Esayoo basalt were also obtained at a distance of 30 m stratigraphically below the sill. Single component magnetizations only were found but the directions are similar to those of the sill and quite different from the Esayoo at greater distances. Except near the margin the magnetization of the Esayoo Formation has not been affected by heat from the intrusion. The sill has not been dated but it is considered to be comagmatic

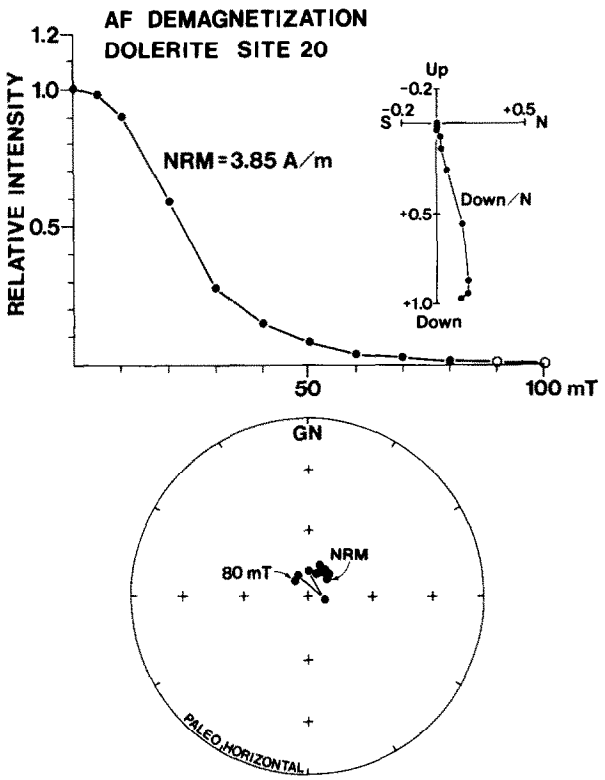


Fig. 7. AF demagnetization of dolerite sill from site 20: single component magnetization. Intensity changes and orthogonal plots above, directions below.

with mid-Cretaceous basalt flows in the vicinity. Its magnetization direction is consistent with that expected for the mid-Cretaceous. Hence the contact test shows that the magnetization of the Esayoo Formation predates the sill and is probably pre-middle Cretaceous.

ANALYSIS AND DISCUSSION

The results are summarized in Table II. The directions at the two localities are in excellent agreement after correction for bedding attitude. This correction has been made by rotation about the strike axis (Fig. 10). Evidently the magnetization predates the tilting of the beds and there has evidently been no relative rotation about inclined axes between the two localities. In Fig. 11 the result is compared to data from cratonic North America. The reference paleopoles for North America are for intervals of 30 Ma duration centered about 270, 260 and 250 Ma. The age range of cratonic data employed in calculating these averages (280–238 Ma) extend just beyond the limits of the Early Permian (approximately 280–250 Ma) within which

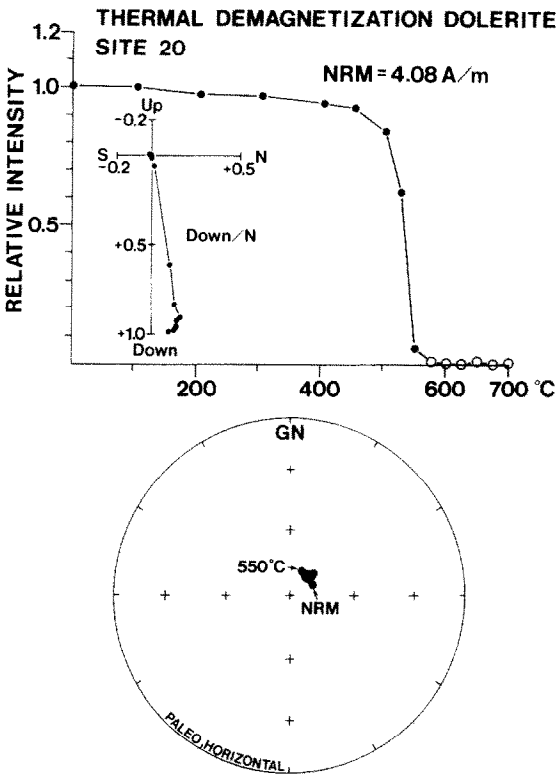


Fig. 8. Thermal demagnetization, dolerite sill, site 20: single component magnetization. Intensity changes and orthogonal plot (one only is given because the direction is near-vertical) above, directions below.

the Esayoo Formation must lie based on the stratigraphic evidence. The north polar distance is consistent with cratonic data but the Esayoo paleopole is situated to the east.

In Table III this comparison is made in another way by calculating the paleolatitudinal displacement and rotation required to explain the divergence of the Esayoo Formation paleopole from the reference paleopoles. None show any significant relative paleolatitudinal displacement, but all show a highly significant sinistral rotation of about 36°. The comprehensive error analysis of Demarest (1983) would yield an error ($P = 0.05$) of 8° (Table III). This result is shown graphically in Fig. 12.

The absence of significant ($P = 0.05$) paleolatitudinal displacement is gratifying for two reasons. Firstly, although there has been substantial tectonic shortening across the Sverdrup Basin evidenced by overthrusts, the amount permitted geologically (certainly not more than 100 km) could not be detected paleomagnetically. It is perhaps worth noting that the mean result does, however, indicate a displacement albeit not significant, in the correct sense—shortening that is and not extension.

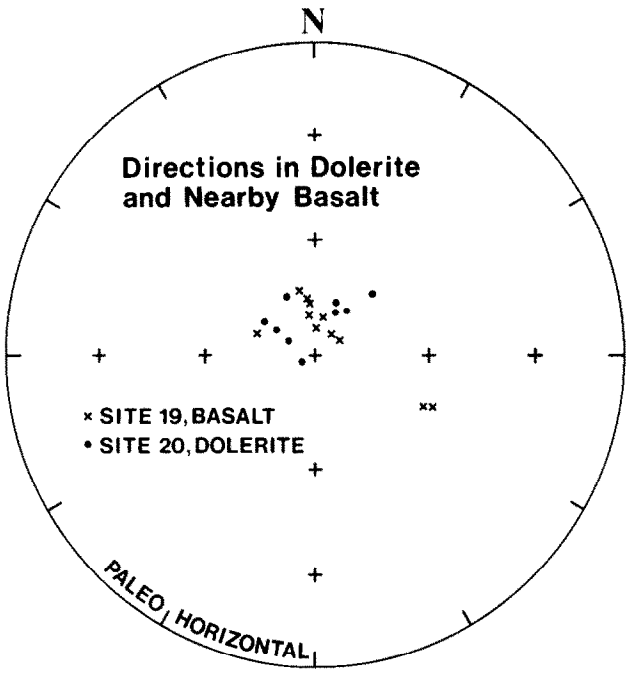


Fig. 9. Direction of magnetization in dolerite sill, site 20, and Esayoo basalt, site 19. The perimeter is the bedding plane of the units into which the sill is intruded.

Secondly, the absence of observable paleolatitudinal displacement confirms, within current accuracies, the paleomagnetic inclinations observed in latest Carboniferous and Early Permian rocks from cratonic North America. This is important because the rocks so far studied in this time interval are all sedimentary

TABLE II

Summary of results

	B	N	Lat. (°N)	Long. (°W)	D_m	I_m	k	α_{95}	Lat. (°N)	Long. (°E)	dm	dp
Van Royen Ridge	7	58	80°52.5′	83°12.5′	128.4	−62.9	106	5.9	49.5	156.5	9.3,	7.3
Hare Fiord	5	30	81°21.0′	81°42.0′	121.6	−60.2	89	8.2	45.2	164.0	12.4,	9.4
Esayoo mean	12	88			126.2	−61.8	105	4.3	47.8	159.0	6.6,	5.1
Mean of poles	12						52	6.1	48.1	159.2		

Notes: B is the number of sites, N the number of specimens. The right-hand columns give the corresponding model A paleopoles with errors (dm, dp). The model B mean of the site poles is given in the last line. Sites given unit weight throughout.

TABLE III

Relative paleolatitudinal displacements (RPD) and relative rotation (RR)

Ma	RPD ^{0}	RR ^{0}
250	-3 ± 9 (7)	38 ± 12 (9)
260	-3 ± 7 (6)	36 ± 10 (8)
270	-5 ± 7 (6)	35 ± 10 (8)

Notes: Displacement (negative denotes displacement from the north) and rotation (positive denotes anticlockwise rotation) have been calculated using the reference mean paleopoles for cratonic North America from Irving and Irving (1982) for the Early Permian. The reference paleopoles are calculated by using a moving window of 30 Ma for the three intervals shown in the left-hand column, and the age range of cratonic data employed is 280 to 238 Ma. Two estimates of error ($P = 0.05$) in RPD and RR are given. The first is by what up to present has been the usual procedure (e.g. Cox, 1980) the second in brackets is by the more comprehensive procedure of Demarest (1983) using his table I.

rocks which are possibly subject to inclination error. The Esayoo basalts, being igneous, are not subject to such errors. Hence we can conclude that inclination errors in the cratonic sedimentary rocks concerned (they are listed in Irving and Irving, 1982, table I entries 69–70) are small, beyond the current resolution of the method. It is not in any case to be expected that errors would be large because their inclinations are low and in a range where inclination errors are small (King, 1955). This result is of some interest because the latest Carboniferous and Early Permian data from cratonic North America are key to the discussion of the fit of continents in the Late Paleozoic (Van de Voo and French, 1974; Irving, 1977; Morel and Irving, 1981; Morel et al., 1981). The results summarized in Table III and Figs. 11 and 12 would appear to discount the possibility of any substantial systematic "hidden" error in the paleomagnetic inclinations, and hence in the paleolatitude of cratonic North America deduced from them.

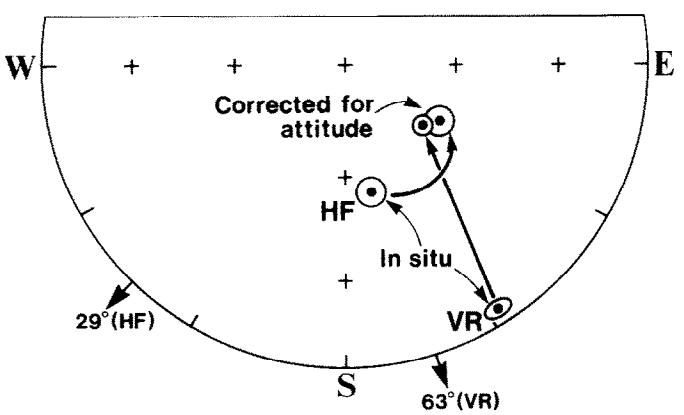


Fig. 10. Corrections for tilt and rotation. *VR* and *HF* are the mean directions at the Van Royen Ridge and Hare Fiord localities and the standard error circles.

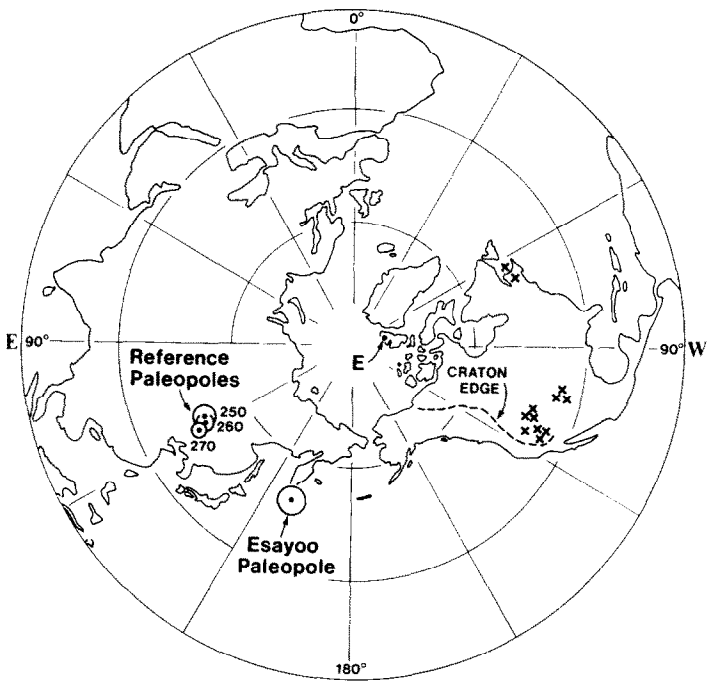


Fig. 11. Comparison of Esayoo paleopole with Early Permian reference paleopoles for North America. Ellesmere Island is indicated by *E*, and Lower Permian localities from which the reference paleopoles were calculated are shown by crosses. Standard errors of paleopoles are given. Further documentation in Table III.

The rotation is the first observed paleomagnetically from the Canadian Arctic Islands. The data tell us that a rotation of about 36° occurred between our sampling localities in the fold and thrust belt of the northern Sverdrup Basin and the reference localities from cratonic North America in eastern Canada and the southwestern U.S.A. (Fig. 11). The data do not tell us where this strain occurred. It could (from our data) have occurred along Lancaster lineament–Maclure Strait; sinistral displacement of 480 km would be implied but seems out of the question on geological grounds (Kerr and Christie, 1965; Kerr, 1977). It could have been formed by the rotation of Greenland relative to cratonic North America as the Labrador Sea and Baffin Bay opened. However, if it is assumed that Ellesmere Island was attached rigidly to Greenland during opening of the Labrador Sea, then it would have been rotated by only 10°, three times less than the 36° observed. More probably the rotation has been produced locally and could be a measure of the motion between the overthrust sheets and the underlying basement. That is, it may have been produced during the formation of the overthrust sheets within the Sverdrup Basin itself, and the rotation observed paleomagnetically is a signature of the style of deformation there. The explanation of the rotation awaits the delineation of the

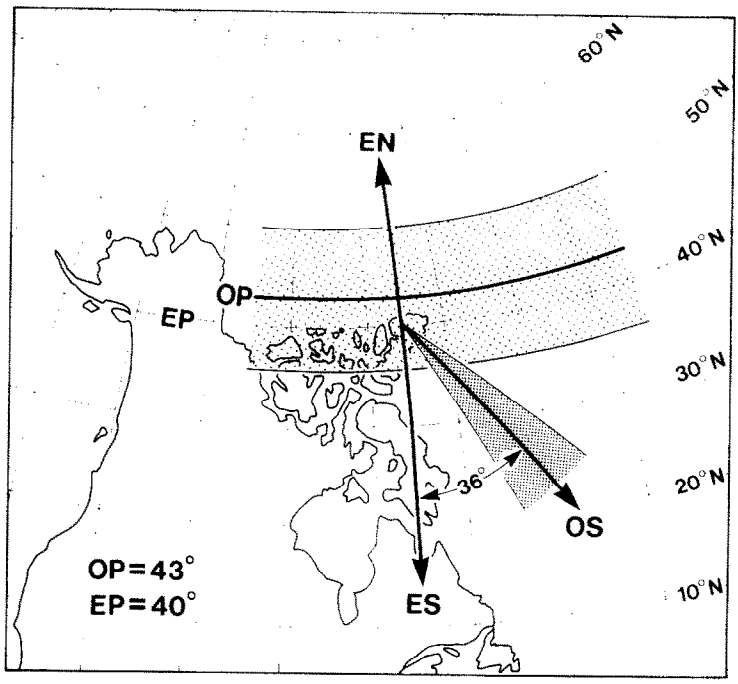


Fig. 12. Paleolatitude and rotation. $EP(=40°N)$ is the expected paleolatitude, $OP(=43°N)$ that observed. *EN* (expected direction of the north pole) and *ES* (expected direction of the south pole) define the expected Early Permian paleomeridian at the sampling localities. *OS* is the observed horizontal component of the mean direction of the Esayoo Formation after correction for geological tilt and denotes the observed direction of paleosouth. The shaded error zones (7° in palcolatitude and 10° in rotation $P = 0.05$) denote uncertainties in the Esayoo determination combined with those in the reference grid. Note that the errors plotted are those calculated by what up till now has been the usual procedure. Application of the more comprehensive procedure of Demarest (1983) would reduce the errors to 6° and 8° respectively (see Table III) narrowing the shaded error zones shown on this figure.

regional extent of such effects (do the rotations persist across the basin, are they variable in amount?). This work is in progress. In meantime therefore we shall confine ourselves to the following brief qualitative remarks.

Geological evidence allows a maximum of about 50 km sinistral shear along Nares Strait between Ellesmere Island and Greenland, far less than the 300 km required by the opening of the Labrador Sea inferred from marine geophysical and other plate tectonic evidence. This is the Nares Strait dilemma which has recently been extensively reviewed (Dawes and Kerr, 1982). As Keen and Peirce (1982) point out one possible way of reconciling this dilemma "is that the Wegener Fault is a fault zone which behaved in a complex manner, rather than a discrete feature as exemplified by the Nares Strait waterway. Several sub-parallel faults may have accommodated the motion at various times, and within this fault zone there may be several fault blocks, some of which may have been rotated. ...At least one minor

strike-slip fault has been observed in eastern Ellesmere Island (Mayr and De Vries, 1982) which could be interpreted as being part of a wide fault zone." The rotation we have observed is in the correct sense (sinistral). However, the fault noted by Mayr and De Vries (1982) is only a short distance from Nares Strait, whereas the rotation we have observed is 200 km northwest of the Strait. The rotation suggests that the deformation has affected a zone at least as wide as this.

If, as seems probable from geological evidence of continuity across Nares Strait, Ellesmere Island has remained, albeit loosely, attached to both Greenland and North America, acting as a bridge between them, then it must have undergone some internal deformation as they moved apart. According to Kristofferson and Talwani (1977) and Srivistava (1978) it is likely to have undergone distributed sinistral shear as well as shortening and compression. The overthrust sheets are evidence of the latter, although geological estimates of shortening to date (Kerr, 1967; Christie et al., 1981; Dawes and Peel, 1981; Dawes and Kerr, 1982b) fall far short of that required from plate reconstructions (Kristofferson and Talwani, 1977; Sclater et al., 1977; Srivistava, 1978). It is possible that the rotation we have observed, although far removed from Nares Strait, could have been produced by an echelon fault system (MacDonald, 1980); it is in the correct sense, and is of a magnitude comparable to rotations observed elsewhere in association with modern complex strike-slip fault systems such as those of New Zealand or California. We must emphasize, however, that the system of structures required to produce the observed rotation has not yet been identified geologically. Nevertheless we feel that the rotation observed provides a valuable clue to the solution of the Nares Strait dilemma.

ACKNOWLEDGMENTS

We wish to express thanks to several people: Dr. W.W. Nassichuk director of the Institute of Sedimentary and Petroleum Geology, G.S.C., Calgary who suggested the investigation and encouraged us greatly; Dr. A. Embry also of that institute who, as party chief, coordinated the field collections in this difficult terrain; and Mr. B. Fischer of the University of Calgary who helped make the collections. The field work was carried out under the auspices of the Polar Continental Shelf Project and we are very grateful to officers of that organization for their help.

REFERENCES

Christie, R.L., Dawes, P.R., Frisch, T., Higgins, A.K., Hurst, J.M., Kerr, J.W. and Peel, J.S., 1981. Geological evidence against major displacement in the Nares Strait. Nature, London, 291: 478–480.

Dankers, P., 1983. Magnetic characteristics of natural hematites at high and low temperatures. Earth Planet. Sci. Lett., in press.

Dawes, P.R. and Kerr, J.W., 1982a. Nares Strait and The Drift of Greenland; A Conflict in Plate Tectonics. Medd. Groenl., Geosci., 8: 1–480.

Dawes, P.R. and Kerr, J.W., 1982b. The case against major displacement along Nares Strait. In: P.R.

Dawes and J.W. Kerr (Editors), Nares Strait and The Drift of Greenland: A Conflict in Plate Tectonics. Medd. Groenl., Geosci., 8: 327–337.

Dawes, P.R. and Peel, J.S., 1981. The northern margin of Greenland from Baffin Bay to the Greenland Sea. In: A.E.M. Nairn, M. Churkin and F.G. Stehli (Editors), The Ocean Basins and Margins, 5. The Arctic Ocean. Plenum Press, London, pp. 201–264.

Demarest, H.H., 1983. Error analysis for paleomagnetic rotation. J. Geophys. Res., 88: 4321–4328.

Irving, E., 1977. Drift of the major continental blocks since the Devonian. Nature, 270: 304–309.

Irving, E. and Irving, G.A., 1982. Apparent polar wander paths Carboniferous through Cenozoic and the assembly of Gondwana. Geophys. Surv., 5: 144–188.

Keen, C.E. and Peirce, J.W., 1982. The geophysical implications of minimal Tertiary motion along Nares Strait. In: P.R. Dawes and J.W. Kerr (Editors), Nares Strait and The Drift of Greenland: A Conflict in Plate Tectonics. Medd. Groenl., Geosci., 8: 327–337.

Kerr, J.W., 1967. Nares submarine rift valley and the relative motion of northern Greenland. Bull. Can. Pet. Geol., 15: 483–520.

Kerr, J.W., 1977. Cornwallis Fold Belt and the mechanism of basement uplift. Can. J. Earth Sci., 14: 1374–1401.

Kerr, J.W. and Christie, R.L., 1965. Tectonic history of Boothia uplift and Cornwallis fold belt, Arctic Canada. Bull. Am. Assoc. Pet. Geol., 49(7): 905–926.

King, R.F., 1955. The remanent magnetism of artificially deposited sediments. Mon. Not. R. Astron. Soc., Geophys. Suppl., 7: 115–134.

Kristofferson, Y. and Talwani, M., 1977. Extinct triple junction south of Greenland and the Tertiary motion of Greenland relative to North America. Geol. Soc. Am. Bull., 88: 1037–1049.

MacDonald, W.D., 1980. Net tectonic rotation, apparent tectonic rotation and the structural tilt correction in paleomagnetic studies. J. Geophys. Res., 85: 3659–3669.

Mayr, U. and De Vries, C.D.S., 1982. Reconnaissance of Tertiary structures along Nares Strait, Ellesmere Island, Canadian Arctic Archipelago. In: P.R. Dawes and J.W. Kerr (Editors), Nares Strait and The Drift of Greenland: A Conflict in Plate Tectonics. Medd. Groenl., Geosci., 8: 167–175.

Morel, P. and Irving, E., 1981. Paleomagnetism and the evolution of Pangea. J. Geophys. Res., 86: 1858–1872.

Morel, P., Irving, E., Daly, L. and Moussine-Pouchkine, A., 1981. Paleomagnetic results from Permian rocks of the northern Saharan Craton and motions of the Moroccan Meseta and Pangea. Earth Planet. Sci. Lett., 55: 65–74.

Sclater, J.G., Hellinger, S. and Tapscott, C., 1977. The paleobathymetry of the Atlantic Ocean from the Jurassic to the present. J. Geol., 85: 509–552.

Srivistava, S.P., 1978. Evolution of the Labrador Sea and its bearing on the early evolution of the North Atlantic. Geophys. J.R. Astron. Soc., 52: 313–357.

Thorsteinsson, R., 1974. Carboniferous and Permian stratigraphy of Axel Heiberg Island, Canadian Arctic Archipelago, Geol. Surv. Can. Bull., 224, 115 pp.

Van de Voo, R. and French, R.B., 1974. Apparent polar wandering for the Atlantic-bordering continents: Late Carboniferous to Eocene. Earth Sci. Rev., 10: 99–119.